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A hybrid encryption model for the hyperspectral images: application to hyperspectral medical images

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Abstract

Hyperspectral images collect information across the electromagnetic spectrum and are widely used to recognize signals, identify materials, and find objects. However, the information in a hyperspectral image may be sensitive and must be protected. Therefore, this paper proposes a new information confidentiality model for hyperspectral images. In this model, Self Adaptive Bald Eagle Search (SABES) optimization algorithm is proposed to select the initial and control parameters of the chaos maps to improve the encryption process. A multi-level chaotic system is implemented to enhance the security of the hyperspectral images by increasing the randomness. A circular shift is utilized to secure the proposed model from differential attack. The proposed encryption model is authenticated with various security analyses. The experimental results show that the proposed cryptographic model is secure from different attacks (statistical, differential, noise, and cropping). The performance of the proposed model is validated by comparing the results of the proposed model with the state-of-the-art methods. The results prove that the proposed model is more secure and requires less computational time for image encryption and decryption in comparison to the existing methods. In addition, the proposed encryption model is applied to secure the hyperspectral medical images, demonstrating its utility in the medical field.

Keywords Chaotic maps · Decryption · Encryption · Hyperspectral images · Metaheuristic optimization algorithm

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1 Introduction

With the meteoric development of the internet, the popularity of multimedia applications, and advancements in computer technology, a large volume of data, including audio, video, text, and images, are transmitted over the internet [20]. Images are an effective way to easily understand information and play an important role in our daily lives [49]. The security of the images is an important issue as the privacy of the person may be affected by unauthorized access. The safe transmission of images requires integrity, confidentiality, authorization, and authentication. Generally, the security of the images can be achieved using three methods: (a) image steganography [4], (b) image watermarking [33], and (c) image encryption [20, 42]. Image steganography embeds secret message in the other images, audio signals, and videos [2]. In this security mechanism, the presence of the secret data is encased into another harmless data or object in a way that only the sender and intended receiver are aware of its presence [3]. Digital watermarking is a non-invasive and content-based technique used for concealing information on computers and with cryptography [53]. It inserts an invisible mark into a document to hide the data inside digital files [14]. Digital watermarking offers a mechanism to add information that is used for the identification of the owners of different rights, track permissions given, and highlights the rights that may be associated with a specific copy or transmission of work [33]. Image encryption methods transmit the images to the receiver after converting the original image into a scrambled image by confusion and diffusion operations [50]. On the receiver side, the decryption procedure is employed by the receivers to retrieve the original image [51]. Image encryption methods are popular, efficient, and the most straightforward method utilized to change the original image into an encrypted form to ensure the security of the images [26]. Different image encryption methods have been presented around the world, but many of these methods are vulnerable to differential and statistical attacks [55]. The traditional encryption techniques such as Blowfish, RSA (Rivest, Shamir, Adleman) algorithm, data encryption algorithm, advanced encryption standard, and international data encryption algorithm, are appropriate for the text encryption. These methods cannot be used for image encryption as images have a large amount of data with high levels of pixel correlation and redundancy. The traditional methods have high processing expense for the image encryption [10, 43, 54]. Therefore, encryption methodologies especially for the images are proposed based on Fourier transformations [11], Rubick cube [58], DNA code sequencing [44], and chaotic maps [28].

Recently, chaos-based encryption methods have become very popular due to the fast, pseudo randomness, deterministic, dynamical, bounded, and aperiodic behavior of chaotic maps [24]. Chaos-based encryption algorithms are the most effective approaches to address the complex, challenging, and intractable issue of fast and highly secure multimedia image encryption [38]. Researchers have proposed different image encryption methodologies based on chaos theory for the commonly used images [31, 45, 49–51]. In digital images, hyperspectral images have unique properties compared to ordinary images and got prominent recognition in the field of military, medical, geological, and mining sectors. A hyperspectral image is a three-dimensional data cube consisting of spectral and spatial information, collected across the electromagnetic spectrum with consecutive narrow spectral bands. As these images contain more sensible and detailed information, there is a need to propose a secure image transmission technique. Thus, in this paper an encryption method based on the multiple chaotic maps for hyperspectral images has been explored.

The rest of the paper is organized as follows. In the Section 2, related work of hyperspectral image encryption methods are presented. The description of the proposed hybrid

encryption model is given in Section 3. Section 4 presents a detailed discussion of the results obtained using the proposed model. The application of the proposed model is explained in Section 5. The last section concludes the work proposed in this paper.

2 Related work

Hyperspectral images are used in a variety of fields, including oceans, agriculture, resources, and the environment. These images contain more information in comparison to the commonly used images. Therefore, the security of hyperspectral images is crucial. Different researchers have proposed chaos-based encryption models for hyperspectral images, as the chaotic maps have the properties such as fast, pseudo randomness, deterministic, dynamical, bounded, and aperiodic behavior [24]. Usama et al. proposed a chaos-based encryption algorithm for the hyperspectral images in which multiple chaotic maps were used to secure the image [38]. Muhaya et al. presented a secure satellite image encryption method with chaotic and Advanced Encryption Standard (AES) algorithms [27]. Villaseñor et al. proposed a multi-chaotic-based encryption algorithm for the hyperspectral images [40]. Geng et al. introduced a hybrid domain encryption algorithm based on composite chaos maps for secure transmission and storage of hyperspectral remote sensing images [15]. Villaseñor et al. proposed a parallel crossed chaotic algorithm for hyperspectral images [41]. Srujana et al. presented a visual secret sharing approach based cryptographic method for the secure transmission of the hyperspectral images. Also, a band selection approach is proposed for removing the redundant spectrum of the hyperspectral images and used in the cryptographic model [35]. Abbas et al. proposed a hybrid encryption method using the Blowfish algorithm and chaotic maps [1].

The existing encryption algorithms use chaotic maps due to their properties, but there are two issues related to chaotic maps. The first is that the chaotic systems are sensitive to their initial conditions and control parameters [19] and the second is that the single chaotic map-based encryption methods are less secure [15]. These issues motivate us to propose an encryption model which is more secure, and have optimal parametric values for the chaotic maps. In this paper, multiple chaotic maps are used for enhancing the security of the hyperspectral images. Initial conditions and the control parameters of the multiple chaotic maps are optimized in the proposed encryption model using Self Adaptive Bald Eagle Search (SABES) optimization algorithm. This hybrid model optimizes the multi-level chaotic maps, thus providing a secure model along with circular shift and random permutation.

3 Proposed hybrid encryption model

This paper proposes a hybrid encryption model for the security of the hyperspectral images. In the proposed model, the SABES algorithm is used for the parameter optimization of the chaotic maps, and optimized parameters are used in the multi-level chaotic maps for the encryption process. The components utilized in the proposed model are SABES optimization algorithm, chaotic maps, random permutation, and circular shift. The layout of the proposed model is given in Fig. 1.

The model works in two modules. In the first module, the spectra of the hyperspectral images are read simultaneously and the SABES algorithm is used for selecting the optimal parametric values of the chaotic maps. The second module starts with the shuffling of image pixel values using a random permutation method to introduce spatial and spectral

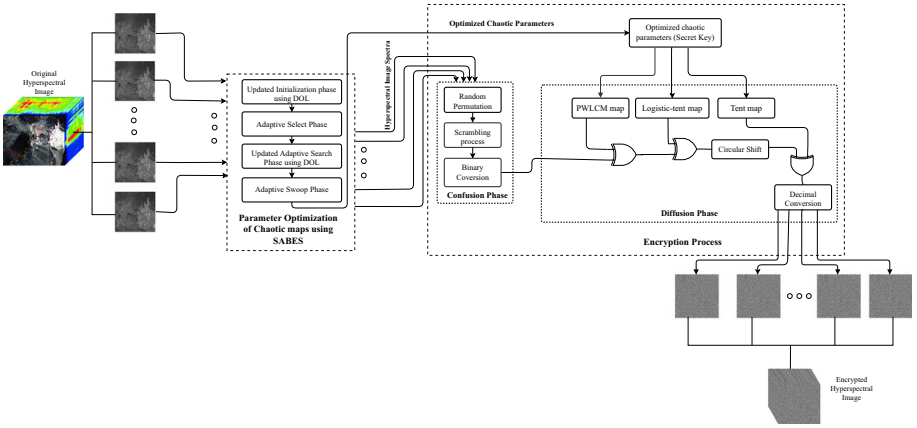


Fig. 1 Layout of the Proposed Hybrid Encryption Model

deformations [7]. The optimal parameters selected in the first module are used in multi-level chaotic maps. In the proposed model, three chaotic maps, i.e. Piecewise Linear Chaotic Map (PWLCM) [47], logistic-tent map [57], and the tent map [37], are used at different levels of the encryption process. In the last stage of the model, the encrypted spectra are collected for the final encrypted hyperspectral image.

The (1) for the PWLCM, (2) for logistic-tent map, and (3) for the tent map, represent mathematical formulation of the chaotic maps:

$$x_i = F(x_{i-1}, \eta_1) = \begin{cases} x_{i-1}/\eta_1, & 0 \le x_{i-1} < \eta_1 \\ (x_{i-1} - \eta_1)/(0.5 - \eta_1), & \eta_1 \le x_{i-1} < 0.5 \\ F(1 - x_{i-1}, \eta_1), & 0.5 \le x_{i-1} < 1 \end{cases} \tag{1}$$

Where the positive control parameter η_1 and initial condition x_i of tent map are $\eta_1 \in [0, 0.5]$ and $x_i \in [0, 1]$.

$$z_{i+1} = \begin{cases} \eta_2 \times z_i(1 - z_i) + (4 - \eta_2) \times z_i/2) \; mod \; 1, & z_i < 0.5 \\ \eta_2 \times z_i(1 - z_i) + (4 - \eta_2) \times (1 - z_i)/2) \; mod \; 1, & z_i \ge 0.5 \end{cases} \tag{2}$$

Where the logistic-tent map parameters $\eta_2 \in [0, 4]$ and $z_i \in [0, 1]$ values.

$$y_i = \begin{cases} \eta_3 \times y_{i-1}, & y_{i-1} < 0.5 \\ \eta_3 \times (1 - y_{i-1}), & y_{i-1} \ge 0.5 \end{cases} \tag{3}$$

Where the tent map control parameter $\eta_3 \in [0, 2]$ and initial condition $y_i \in [0, 1]$.

In Fig. 2, the bifurcation diagrams of the chaotic maps are shown. From the figure, different chaotic behaviors of the maps can be seen. Using the PWLCM map, the input values are divided into three linear segments that have particular limits. But, PWLCM map does not produce discrete uniform output for discrete uniform input [36]. Therefore, after making the segments, the logistic-tent map is used in the second stage for uniform distribution of the output segments [6]. In the end, the tent map is applied to range the dynamic behavior of the encrypted segments [52]. Thus, in the proposed model, randomness in the pixel values

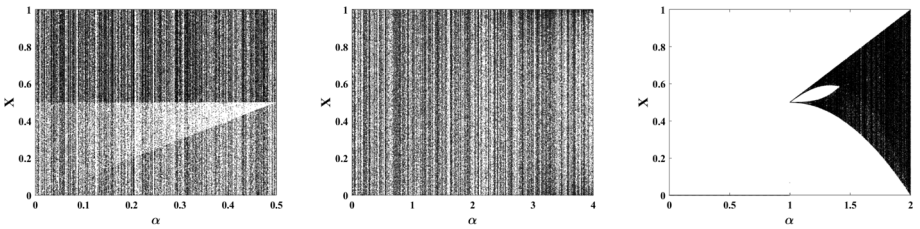


Fig. 2 Bifurcation diagrams of (a) PWLCM map, (b) Logistic-Tent map, (c) Tent map

is increased using multiple chaotic maps applied at different levels to improve the security. A bit-wise circular shift [31] is used to increase defense against the differential attacks.

The proposed encryption model has two modules; parameter optimization and encryption. Further, the encryption module is divided into two phases: confusion and diffusion. A detailed description of the proposed model is given in the following subsections.

3.1 Parameter optimization

In this module, the SABES optimization algorithm is used to optimize the initial conditions and control parameters of the chaotic maps. A detailed description of the process is given below:

3.1.1 Bald eagle search optimization algorithm

Alsattar et al. presented a Bald Eagle Search (BES) optimization algorithm in 2020, based on the bald eagles' food source searching and hunting behavior [5]. In this technique, bald eagles are used as search agents to find the optimal solution to the problems. The operation of the BES algorithm is divided into three phases: selecting a space from the entire search space (exploration), the search phase (exploitation), and the swooping phase. The explanation of these phases is given below:

- **Selecting the space from the entire search space:** The best area is chosen from the previously provided search stage information in this phase using the following formula:

$$SP_{i,new} = SP_{best} - \alpha \times r(SP_{mean} - SP_i) \tag{4}$$

  Where i is the number of the search agents, r is a random variable that takes a value between [0, 1], and α is in the range [1.5, 2]. Using the values of SP_{best} and SP_{mean}, the search agents randomly find a new region closer to the previous search space. The SP_{best} values represent the search agents' best position for hunting in the previously defined search area. SP_{mean}, on the other hand, represents the average of all the eagle positions from the prior points.
- **Search phase:** In the search phase, agents detect the prey in the designated region selected in the first phase. The following mathematical expression is used to update the

position:

$$
\begin{aligned}
SP_{i,new} &= SP_i + m(i) \times (SP_i - SP_{i+1}) + l(i) \times (SP_i - SP_{mean}) \\
where, l(i) &= \frac{lr(i)}{max(|lr|)} \quad and \quad m(i) = \frac{mr(i)}{max(|mr|)} \\
lr(i) &= r(i) \times \sin(\theta(i)), \quad mr(i) = r(i) \times \cos(\theta(i)) \\
\theta(i) &= \alpha \times \pi \times rand, \quad r(i) = \theta(i) + R \times rand
\end{aligned} \tag{5}
$$

The range of α is 5 to 10, $rand$ is a random variable with values ranging from [0, 1], and the number of search cycles is represented by the R, which has a range of [0.5, 2].

- **Swooping phase:** The search agents choose the optimal position to find the best solution in the swooping phase. The following is an illustration of the formulation:

$$
\begin{aligned}
SP_{i,new} &= rand \times SP_{best} + l1(i) \times (SP_i - c1 \times SP_{mean}) \\
&\quad + m1(i) \times (SP_i - c2 \times SP_{best}) \\
l1(i) &= \frac{lr(i)}{max(|lr|)} \quad and \quad m1(i) = \frac{mr(i)}{max(|mr|)} \\
lr(i) &= r(i) \times \sinh[\theta(i)], \quad mr(i) = r(i) \times \cosh[\theta(i)] \\
\theta(i) &= \alpha \times \pi \times rand, \quad r(i) = \theta(i)
\end{aligned} \tag{6}
$$

where $c1$, $c2$ are the random number having a range between 1 and 2.

3.1.2 Proposed multi chaotic maps parameter optimization using SABES algorithm

In the Bald Eagle Search optimization algorithm, the algorithmic parameters are α, R, c_1, and c_2. The researchers choose the values of these algorithmic parameters after assessing the various values of parameters within the prescribed constraints. However, it is likely that the chosen values are only appropriate for the specific issues. Also, it may have the problem of local optima stagnation and premature convergence. This work proposes optimization of the Initial conditions and the control parameters of the multiple chaotic maps using an improved version of the BES algorithm. The following are the improvements:

- The time-varying adaptation mechanism is used to select the value of the algorithmic parameters of the BES algorithm
- Dynamic-opposite learning is applied in the proposed algorithm throughout the initialization and search phase of the bald eagle search optimization algorithm.

The enhancements are produced by employing dynamic-opposite learning to improve the search abilities of the optimization. The BES algorithm is improved to self adaptive BES and is applied to optimize the parameters of the chaotic maps. The specific changes are as follows:

- **Time-varying adaption mechanism of SABES:** The algorithmic parameter values have a significant impact on the performance of metaheuristic optimization algorithms. The optimization algorithm uses algorithmic parameters to manage the search process. The BES algorithm has four algorithmic parameters (α, R, c_1, and c_2). The values of these parameters are chosen in the BES algorithm on experiment basis. To find the values of these parameters, researchers have utilized a variety of approaches, the most common of which is the time-varying adaptation mechanism. The algorithmic parameters are chosen using linear [12] and non-linear [9] time-varying adaptation mechanism.

Equations (7) and (8) define the linear and non-linear adaptation strategies, respectively.

$$v(c_{itr}) = \frac{itr_{max} - c_{itr}}{itr_{max}}(v_{max} - v_{min}) + v_{min} \tag{7}$$

$$v(c_{itr}) = \frac{(itr_{max} - c_{itr})^n}{(itr_{max})^n}(v_{max} - v_{min}) + v_{min} \tag{8}$$

Where itr_{max} denotes the maximum number of iterations, c_{itr} represents the current iteration, v_{max} denotes the upper limit of the algorithmic parameter, and v_{min} is the lower limit of the algorithmic parameter.

The SABES algorithm uses a non-linear adaptation mechanism in the select and search phases to improve the exploration and exploitation searching capabilities. The linear adaptation approach is utilized to determine the overall optimal solution and to prevent local optima in the swoop phase.

- **Initialization and Search phase updation using Dynamic-opposite learning method:** The most successful learning techniques are opposition-based learning approaches, and the goal is to improve the search capability [32]. The core principle is that opposing candidate solutions are more likely to be nearer to the optimal solution than randomly produced solutions and more advantageous for producing new solutions throughout the search phases [29, 39]. Various opposition strategies are presented and combined with the algorithms to improve their search powers [13, 17, 21, 25, 30]. However, most of the opposition-based approaches use deterministic methods to identify opposing candidate solutions, which may not be adequate to improve exploitation capabilities. Xu et al. [48] proposed Dynamic-Opposite Learning (DOL) to address the shortcomings of the opposition-based learning, quasi-opposite-based learning, and quasi-reflection-based learning approaches. The DOL's mathematical equation is as follows:

$$\begin{aligned} P^{DO} &= P + \omega \times r \times (r \times P^0 - P) \\ P^0 &= a_j + b_j - P_j, \quad j = 1, 2, ..., D \end{aligned} \tag{9}$$

Where a and b represent the problem's lower and upper borders, respectively, and D represents the problem dimension. r is a random variable in the range [0, 1]. P is a real number representing search agent locations and has values ranging from a to b. The opposition learning approach is used to acquire P^0 values. ω is the weighting factor with a value greater than zero.

Dynamic-opposite learning is used in the SABES algorithm to improve the initialization of search agent locations. The initiation step influences the convergence speed and the process of finding the best solution. As a result, DOL has been used in the proposed algorithm to boost the likelihood of convergence towards the global optimum while avoiding the local optimum stagnation problem. The search agents' random locations were generated using a uniform random distribution approach as given below:

$$P_{i,j} = a_j + (b_j - a_j) \times rand \tag{10}$$

where $P_{i,j}$ symbolizes the location of the ith search agent created at random ($i = 1, 2, ..., M$: search agents number) and j represents the dimensions of problem variables ($j = 1, 2, 3, ..., D$: problem variables). The problem's upper and lower limits are a and b. $rand$ represents a random variable with a value between [0, 1]. After the search agent positions have been created at random using (10), dynamic-opposite learning locations have been generated using (9). The fitness values for the created locations

are then calculated. M search agents with fittest values are chosen at random from the positions set (P) and the DOL set (P^{DO}) and employed for further computations.

In addition, dynamic-opposite learning has been used to accelerate the convergence rate by balancing the search phases. Dynamic-opposite learning is used in the search phase (exploitation) to seek a better candidate solution by simultaneously considering the candidate solutions and their corresponding opposite candidate solutions, hence expanding the exploitation space.

The flow chart of the proposed multi chaotic maps parameter optimization using SABES algorithm is presented in Fig. 3.

3.2 Encryption process

The encryption process starts with the confusion phase, in which a random permutation is used to change the spatial and spectral data shuffling without changing the actual values of the data. After the shuffling process, the spatial pixel values are converted into binary values.

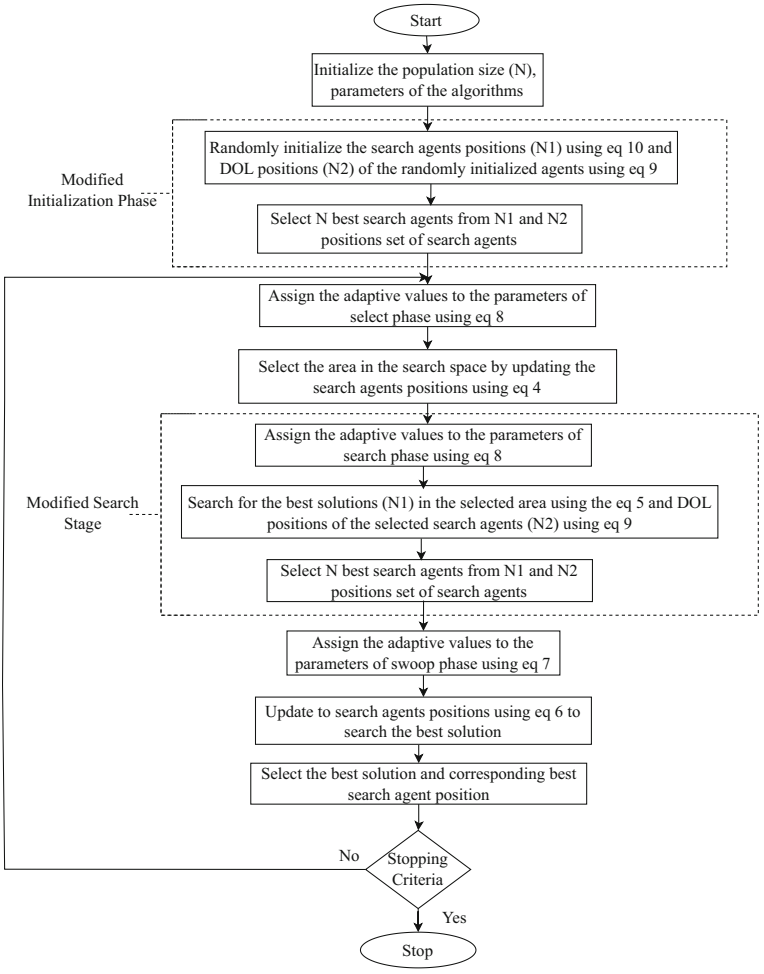


Fig. 3 Proposed chaotic maps parameter optimization using SABES algorithm

In the next step, three chaotic maps are utilized in three levels of the diffusion process. In the first level, the scrambled input pixel values (p_1) are XORed with the optimized PWLCM data, and new values are considered as (p_2). In the next level, optimized logistic-tent map data values are XORed with the data values (p_2). Then, a bit-wise circular shift operation is performed on the newly generated values (p_3) of the second level to produce new random values (p_4). In the last level, data values (p_4) are XORed with the tent map to build the final encrypted values (d_5). At the end of the encryption process, the encrypted binary data values are converted into decimal data values.

The decryption procedure is exact opposite of the encryption process. In the first step, the encrypted image is converted into binary values, and then the XOR operation is performed with the tent map values obtained using the secret key used in the encryption process. The bit-wise reverse circular shift is applied to revert the bits. The logistic-tent map values are attained by employing the secret key. The XORed operation is applied to the reverted bits and the logistic-tent map values. At the last chaotic level, the XOR operation is performed on the data values, and PWLCM map values generated using the secret key. In the end, the reverse scrambling process is applied to generate the spatial and spectral layers of the data to obtain the hyperspectral image.

4 Results and discussions

The experiments of the proposed hybrid encryption model are conducted in the MATLAB 2021a, on Windows 10 Pro operating system with Intel Xeon Silver 4215 (2.50 GHz dual processors) and 256 GB of RAM. The parameters setting used in this experiment are population size: 30 and the number of iterations: 100. Thirty-two hyperspectral images are used for performance evaluation of the proposed cryptographic model. The dataset description is represented in Table 1.

The results of the proposed hybrid cryptographic model have been analysed using visual, histogram, entropy, NPCR, UACI, pixel similarity, pixel dissimilarity, randomness and execution time analysis.

4.1 Visual analysis

The visual analysis is used for the determination of no information leakage while analyzing with the naked eyes. In Fig. 4, the original, encrypted and decrypted hyperspectral, grayscale and RGB images are shown to inspect the performance of proposed model.

Figure 4, depicts that no information about the original images can be gleaned by gazing at encrypted images. Additionally, it has been noted that the original and the decrypted images are exactly the same. This suggests that applying the proposed model does not result in any information being leak from the original images.

4.2 Histogram analysis

The histogram represents an image's pixel value distribution for different intensity or brightness levels. The image is secured from the intruders by converting the pixel distribution information with uniform distribution for an image. The 2D and 3D histogram analysis of the hyperspectral images are shown in Fig. 5. In the 2D histogram plot, the pixel intensity values of a single layer of the hyperspectral image are shown. In the 3D histogram, the intensity values of the complete hyperspectral image are shown (Fig. 6).

Table 1 Dataset Description

Dataset Name	Dimension
Indian Pines	145 × 145 × 220
Pavia University	1096 × 715 × 102
Kennedy Space Center (KSC)	512 × 614 × 176
Botswana	1476 × 256 × 145
Salinas	512 × 217 × 224
China Change Dataset (T1)	420 × 140 × 154
China Change Dataset (T2)	420 × 140 × 154
USA Change Dataset (T1)	307 × 241 × 154
USA Change Dataset (T2)	307 × 241 × 154
Cuprite	512 × 614 × 224
Washington DC	1280 × 307 × 191
Samson	95 × 95 × 156
Selftestrad	280 × 800 × 126
Selftestrfl	280 × 800 × 126
Urban 1	307 × 307 × 162
Urban 2	307 × 307 × 210
Jasper Ridge	100 × 100 × 198
IEEE Grss CASI	349 × 1905 × 149
Sandiego	256 × 256 × 189
Abu-Airport-1	100 × 100 × 205
Abu-Airport-2	100 × 100 × 205
Abu-Airport-3	100 × 100 × 205
Abu-Airport-4	100 × 100 × 291
Abu-Beach-1	150 × 150 × 188
Abu-Beach-2	100 × 100 × 193
Abu-Beach-3	100 × 100 × 198
Abu-Beach-4	150 × 150 × 162
Abu-Urban-1	100 × 100 × 204
Abu-Urban-2	100 × 100 × 207
Abu-Urban-3	100 × 100 × 191
Abu-Urban-4	100 × 100 × 205
Abu-Urban-5	100 × 100 × 205

4.3 Analysis of entropy, NPCR, and UACI

An entropy value indicates the presence of randomness created in the encrypted image using the encryption method. The higher is the entropy value, the more secure is the image from the entropy attacks. The values of Number of Pixels Change Rate (NPCR) and Unified Average Changing Intensity (UACI) indicate the security of the encryption algorithm against the differential attacks [46]. These values are used to represent that even after one pixel change in the original image, different encrypted images are created using the same encryption algorithm. NPCR and UACI values indicate the quantitative and qualitative analysis. The metric values calculated for five hyperspectral image using the proposed hybrid

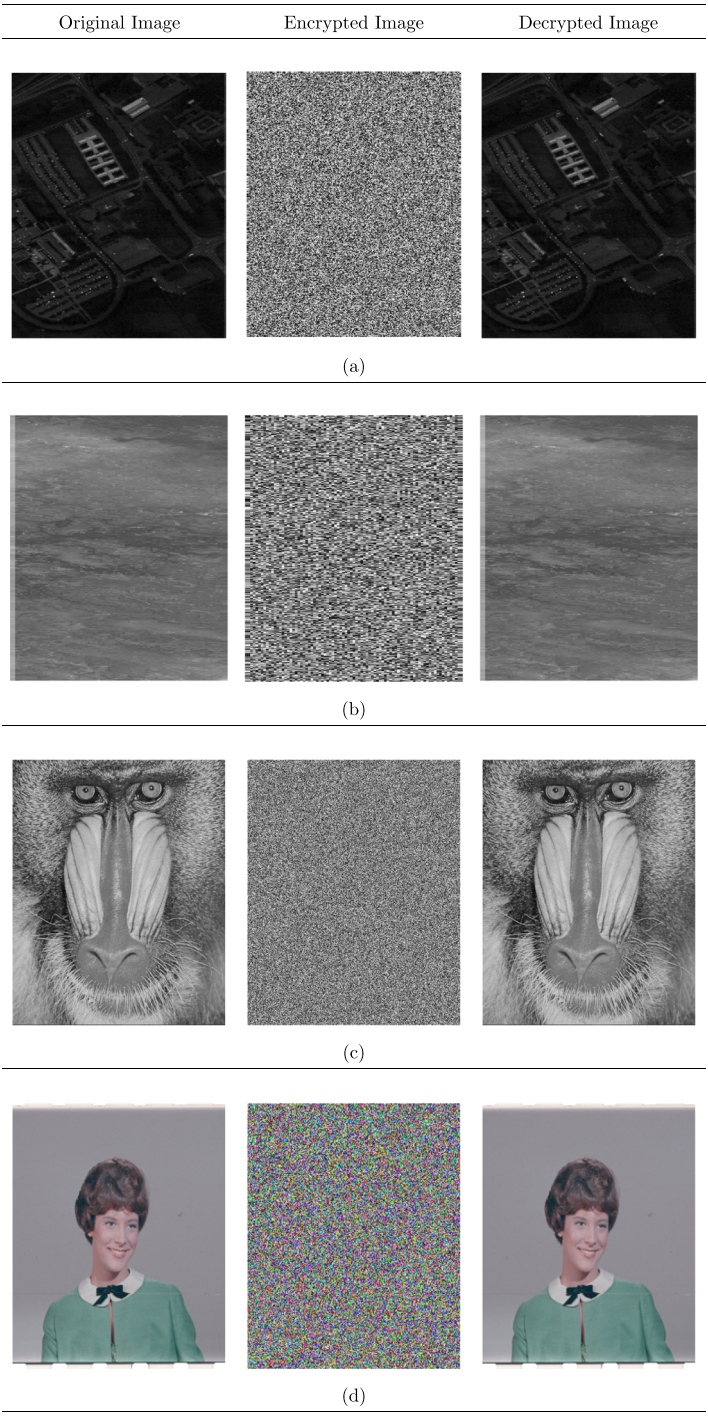


Fig. 4 Visual analysis of encrypted images (a) 4th Spectra of Pavia University hyperspectral image, (b) 4th Spectra of Botswana hyperspectral image, (c) Madrill grayscale image, and (d) Female RGB image

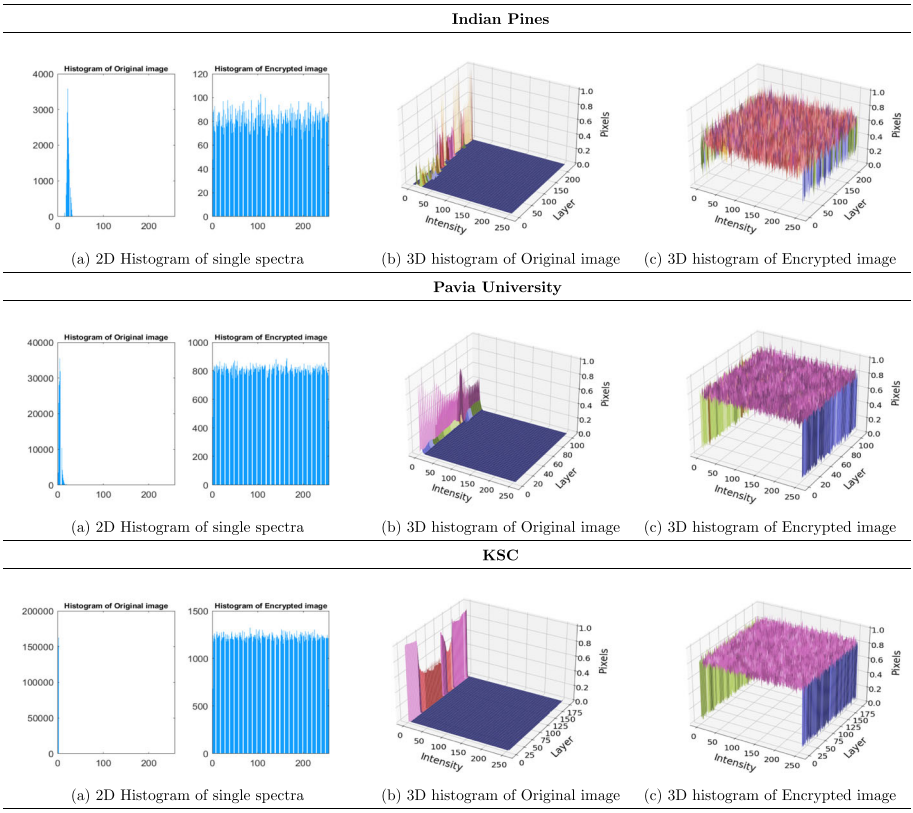


Fig. 5 2D and 3D Histograms of Satellite hyperspectral images

model are presented in Table 2. The evaluation results for rest of the hyperspectral image are presented in Appendix A.

From Table 2, it has been observed that the entropy values are close to 8, which indicates that the proposed hybrid encryption model is secure from entropy attacks. The proposed model security against differential attacks has been proved as the obtained NPCR values are nearer to one and UACI values are lower. The results prove that the proposed hybrid cryptographic model is reliable, efficient, and secure.

4.4 Analysis of pixel similarity and dissimilarity metrics

Pixel Similarity Metrics: The pixel similarity metrics are used to analyze the relationship between the structure of the original and encrypted hyperspectral image. In order to estimate the structural dissimilarity after encryption, the various similarity measures that may exist between original and encrypted images have been computed. Thus, Structural Similarity Index (SSIM), Structural Content (SC) [8], Normalized Cross-Correlation (NCC) [23], Image Fidelity (IF) [34] metrics are used to analyse the variations in the structure of original and encrypted hyperspectral images. The SSIM is used to examine the structural similarity between the original and decrypted image [22]. Also, the SSIM is used to analyse the divergence, luminance, and assembly between the original and encrypted hyperspectral image. NCC is employed to analyze the correlation resemblance between the images. The

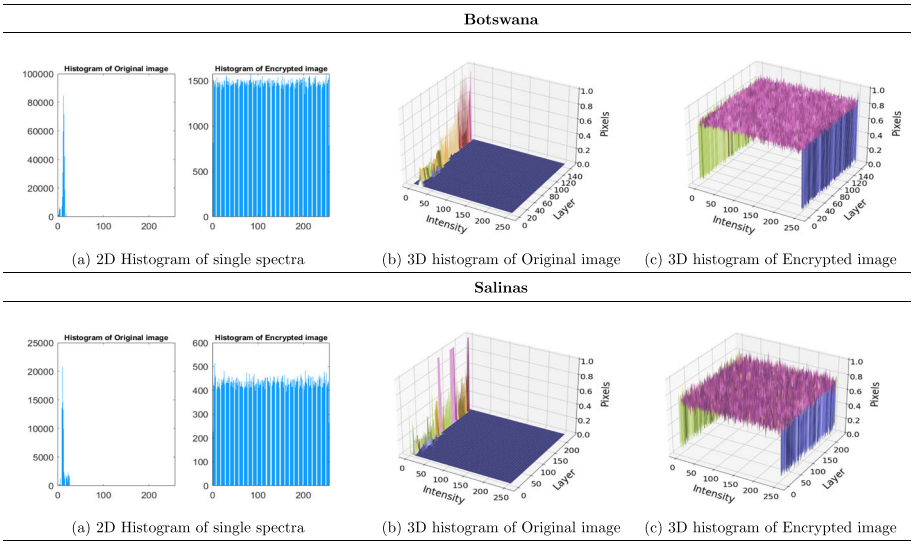


Fig. 6 2D and 3D Histograms of Satellite hyperspectral images

sharpness and the noise levels present in the images are measure using the SC metric. IF metric computes the imperceptibility between the original and the encrypted hyperspectral images. The highest level of resemblance between consecutive pixels in an image may be close to 1, whereas the highest degree of dissimilarity between nearby pixels will provide a similarity value that is close to 0.

Pixel Dissimilarity Metrics: The error present in the original and the encrypted images are calculated using the pixel dissimilarity metrics. The Peak Signal to Noise Ratio (PSNR) [22], Mean Squared Error (MSE) [22] and Mean Absolute Error (MAE) metrics are used to examine the variations in the original and the encrypted images. The PSNR values depict the noise effect in the original and the encrypted image. The lower values of PSNR show the high dissimilarity between the input images due to the high noise availability. The MSE reflects the total squared variation between the original and encrypted images. Whereas, MAE is used to calculate the absolute error between the input images.

In Table 3, the values achieved for the pixel similarity and dissimilarity metrics are shown for the five encrypted hyperspectral images to check the robustness of the proposed hybrid model. The analysis of the remaining images are given in Appendix B.

Table 3 shows that the SSIM value is 1 for the original image vs. decrypted image, which indicates that the decrypted image obtained from the proposed decryption model has a 100% similar structure. In contrast, the SSIM values of the original and encrypted images are close to 0, which shows the structural dissimilarity between the original and encrypted images. Also, the values attain for the pixels of the encrypted images show that the encrypted hyperspectral images are very different from the original images. Thus, the original image is very secure from the intruders.

4.5 Randomness analysis

The randomness of the numbers generated by a chaotic system should be examined using different tests before using the encryption process. The popular standard tests like the

Table 2 Evaluation metrics for a comparison between original, encrypted, and decrypted Satellite images

Image Name	Entropy		NPCR	UACI
	Original	Encrypted		
Indian Pines	3.8190	7.9990	1	0.3334
Pavia University	3.6689	7.9987	1	0.3334
KSC	0.9017	7.9987	1	0.3334
Botswana	4.0704	7.9986	1	0.3339
Salinas	3.6402	7.9988	1	0.3339

TestU01, Diehard, and NIST tests are widely used for the analysis. In this work, NIST 800-22 test suit is used for the randomness analysis.

There are a total of fifteen tests in the NIST 800-22 test suit which are recommended by NIST US. For each chaotic system utilized in the proposed model, 100 series with a stream length of (i=100000 bits) are used to run all these statistical tests. The approval of this test is based on the p-value. The test is analysed at the confidence interval of $\alpha = 0.01$. If the resulting p-value in each test is greater than zero, the resulting sequence is random and successfully passes the test with a confidence level of 99%. Additionally, the generated sequence passes the 15 NIST tests, if the success percentage is higher than 96%. The result is presented in Table 4.

Table 4 indicates that the chaotic maps used in the proposed model have successfully passed the NIST test which shows that the proposed algorithm has required random behavior.

4.6 Analysis of noise attacks

The encryption algorithms should be resistant to data loss threats during data storage and communication [18]. In addition, noise disruption is unavoidable in real-time applications. The encryption method should withstand noise attacks. The proposed model efficiency is checked by adding different types of noises in the encrypted image. Figure 7 shows the effectiveness of the proposed model against noise attacks.

From Fig. 7, it has been observed that the proposed model is quite resistible against different types of noise attacks. Thus, the proposed model is efficient for real-time applications.

4.7 Analysis of cropping attack

The anti-shearing ability of the propounded encryption algorithm is verified using the cropping or occlusion attack [42]. The robustness of the proposed model against the cropping attacks is presented in Fig. 8.

The encrypted image is presented on the left side of this figure, no visual information about the original image is visible. The encrypted image presented in the figure with no cropping, vertically cropping, horizontally cropping, and block cropping. In Fig. 8, (a) encrypted image is decrypted using the secret key, and the decrypted image is recovered with no loss. In (b) portion of the image, the encrypted image is cropped vertically, and with the same keys, the image is recovered with only the loss of the pixel values which are corrupted during the cropping attack. The same scenario can be seen in the (c) and (d) portions of the figure, in which the image is cropped horizontally and in the block shape.

Table 3 Result analysis of the proposed hybrid encryption model using pixel similarity and dissimilarity metrics

Image Name	Pixel Similarity Metrics					Pixel Dissimilarity Metrics		
	SSIM		SC	NCC	IF	PSNR	MSE	MAE
	Original vs. Encrypted	Original vs. Decrypted						
Indian Pines	0.0016	1	0.0061	0.8600	-14.5613	5.2673	1.28E+09	60.9013
Pavia University	0.0009	1	0.0019	0.7510	-28.5098	5.0474	1.34E+09	20.8987
KSC	0.0001	1	0.0059	0.3588	-194.6945	4.7910	1.43E+09	64.5571
Botswana	0.0011	1	0.0037	0.8296	-22.0660	5.1151	1.32E+09	39.9217
Salinas	0.0007	1	0.0017	0.7781	-37.1495	4.9851	1.36E+09	18.2475

Table 4 NIST test results for the chaotic maps used in proposed hybrid encryption model

Test Name	PWLCM Map			Logistic-Tent Map			Tent Map		
	p-value	Success Proportion	State	p-value	Success Proportion	State	p-value	Success Proportion	State
Frequency	0.534146	100%	Success	0.911413	100%	Success	0.350485	100%	Success
Block Frequency	0.911413	100%	Success	0.213309	100%	Success	0.739918	100%	Success
Runs	0.350485	100%	Success	0.739918	100%	Success	0.911413	100%	Success
Longest Run (M=1000, N=100)	0.534146	100%	Success	0.991468	100%	Success	0.739918	100%	Success
Rank	0.350485	100%	Success	0.911413	100%	Success	0.911413	100%	Success
FFT	0.739918	100%	Success	0.08879	100%	Success	0.122325	100%	Success
Non-Overlapping Template (m=9)	0.911413	100%	Success	0.739918	100%	Success	0.534146	100%	Success
Overlapping Template (m=9)	0.66882	100%	Success	0.534146	100%	Success	0.350485	100%	Success
Universal	0.92379	100%	Success	0.917157	100%	Success	0.928157	100%	Success
Linear Complexity	0.534146	100%	Success	0.213309	100%	Success	0.911413	100%	Success
Serial	0.350485	100%	Success	0.534146	100%	Success	0.739918	100%	Success
Approximate Entropy	0.350485	100%	Success	0.350485	100%	Success	0.122325	100%	Success
Cumulative Sums	0.739918	100%	Success	0.911413	100%	Success	0.534146	100%	Success
Random Excursions (x= -4)	0.596514	100%	Success	0.92359	100%	Success	0.882363	100%	Success
Random Excursions Variant (x= -9)	0.322475	100%	Success	0.944824	100%	Success	0.746069	100%	Success

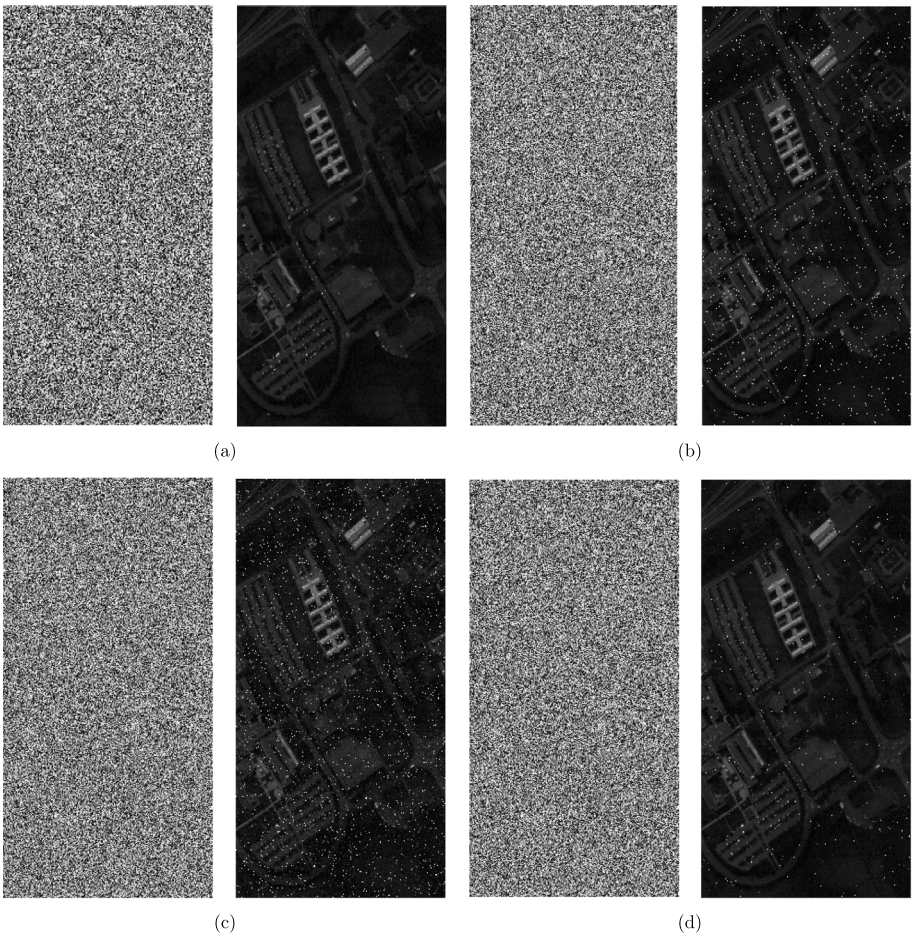


Fig. 7 Noise attack analysis of Pavia University encrypted and decrypted image after (a) no noise (b) Salt and Pepper noise (c) Gaussian noise (d) Speckle noise

4.8 Key sensitivity analysis

The keys used in the decryption process to fetch the original image play an important role in the cryptographic models [31]. A small change in the key should affect the resultant image [7]. In the proposed model, six keys are used for the image encryption process. These keys are required in the decryption process, and even small change in the key-value will distort the image. In Table 5, the original and the modified key-values are presented. In the modified key set, only the first key-value is changed with a small value (10^{-16}) (The bold and underlined value at 16th digit after the decimal point in the table indicates the change of one digit). The resultant decrypted images computed using the original keys set and modified keys set are depicted in Fig. 9.

From Fig. 9, it has been observed that the original image and decrypted image computed using the original keys set are exactly the same. Whereas the decrypted image fetched using the modified keys set has no similarity with the original image. This indicates that the proposed model is very sensitive to their keys values.

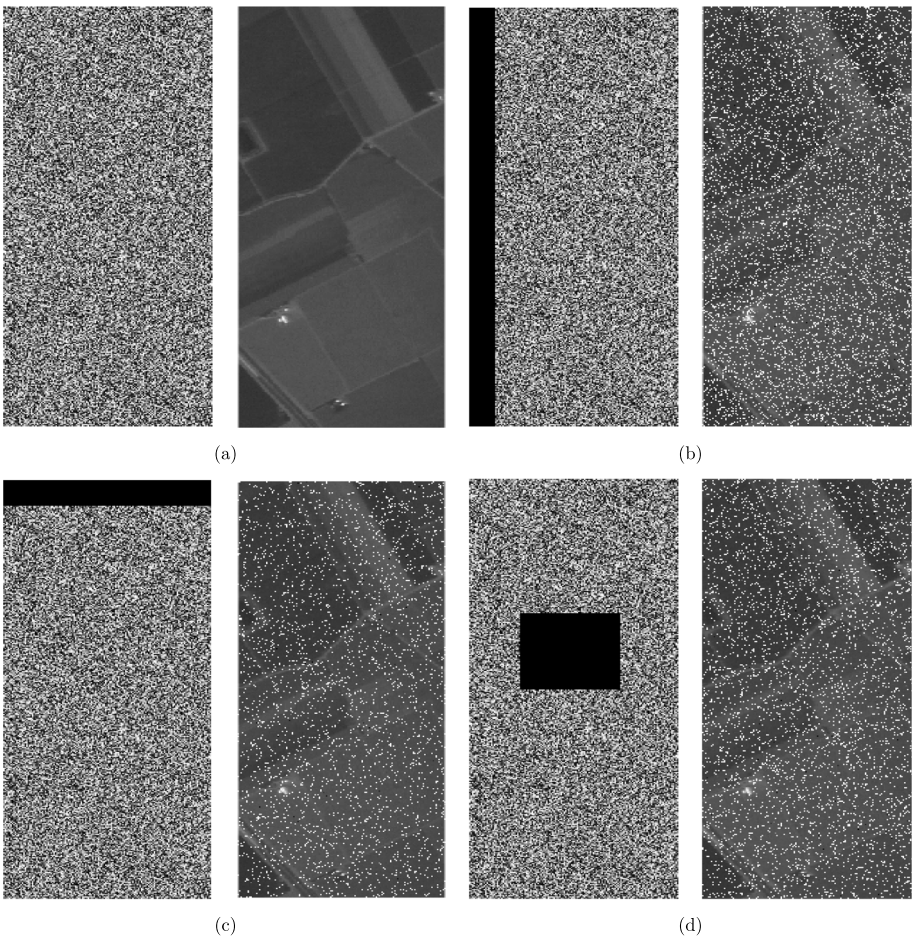


Fig. 8 Cropping attack analysis of Salinas encrypted and decrypted image after (a) no cropping, (b) vertical cropping, (c) horizontal cropping and (d) block cropping

4.9 Key space analysis

Key space analysis is an essential technique that is used to determine the feasibility of a random sequence generator to produce encryption keys. A larger key space ensures the security

Table 5 Analysis of key-values for the Botswana hyperspectral image

Key Number	Original key-value	Modification key-values
K1	0.2909345693667922	0.2909345693667923
K2	0.2659041218126361	0.2659041218126361
K3	0.2930631278957424	0.2930631278957424
K4	0.3912829310439881	0.3912829310439881
K5	0.2485327387386953	0.2485327387386953
K6	0.6192089984178624	0.6192089984178624

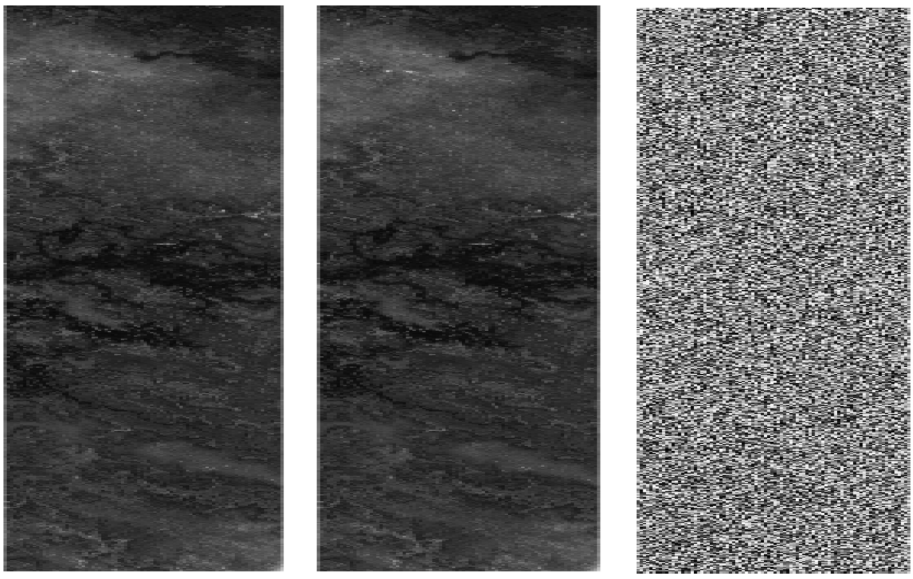


Fig. 9 Original image (a) and decrypted images of Botswana using (b) correct key-values (c) modified key-values

from the brute force attack. The encryption algorithm's key space should be large enough to assure the security of the image [42]. Six keys have been employed in the proposed hybrid encryption model, with keys representing chaotic maps' control parameters and initial values. The proposed model uses six keys with a precision of 10^{16}, and the total key space used is $10^{(16\times 6)} = 10^{96}$, which is much higher than 2^{128}. Thus, the proposed model has enough key space to endure severe attacks and can resist brute force attack.

4.10 Computational time analysis

The execution time of the Cuprite hyperspectral image which has 224 layers, 512 rows, and 614 columns is analyzed to prove the efficiency of the proposed model. For the computation time analysis the results are taken from [15]. Table 6 represents the execution time comparison of the propounded hybrid encryption model with the existing encryption methods. The computation time of remaining images is available in Appendix A. From the table, it can be seen that the proposed model much less computational time for both encryption and decryption of the hyperspectral image in comparison to the existing methods.

Table 6 Encryption and decryption time consumption for Cuprite hyperspectral image

Cryptographic Method	Encryption Time (s)	Decryption Time (s)
Logistic Map [15]	66.24	66.21
Chebyshev Map [15]	66.17	66.05
AES-256 [15]	2480.21	2488.13
Hybrid Domain Encryption [15]	67.96	67.74
Proposed Model	32.4078	22.8544

Also, the comparison of the vertical and horizontal correlation coefficient values of encrypted hyperspectral image, key space, entropy values, achieved using the different cryptographic methods for the Cuprite hyperspectral image is presented in Table 7.

From the table, it has been observed that the proposed hybrid encryption model has attained better results in comparison to the state-of-the-art method.

5 Application of the proposed model to the hyperspectral medical images

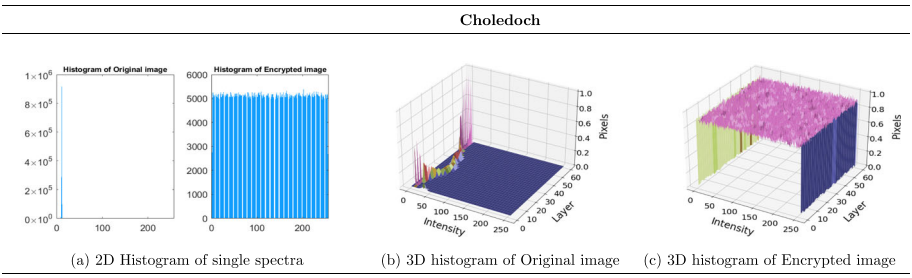
With the adoption of digitization in the medical field the electronic medical data can be easily distributed among the doctors and the respective patients. The electronic medical data includes recommended dosages for medicines, clinical medical records, information about insurance, and other personal data. Thus, electronic medical data contains sensitive information of the patients and medical data stealing can have worse impact on the patients. In the medical data, medical images have major portion as images play an important role in medical diagnosis. Also, medical images contain a major portion of the data transfer during telemedicine services. Telemedicine offers medical services remotely using physiological monitoring devices and telecommunications systems. This scheme provides healthcare delivery services remotely to the patients when patients are unable to reach or access the required medical services [16]. The recent advancements in information and telecommunication technologies have created more convenient methods for telemedicine services. However, the medical data shared over the internet have a probable security risks to patient information transmission over insecure networks. Hence, secure authentication and high confidentiality schemes are required.

In this paper, the proposed model is applied to medical hyperspectral images to ensure the security of the images. As, these images contain more information in comparison to the other images. These images have high information content and large number of layers; thus, these images require encryption model which provides high security with low computation time. The proposed model has fulfilled the need of low computation time for encryption and decryption with high level of the security.

In this work, a medical hyperspectral image is used to validate the proposed model for encryption of the images. The dataset that has been used for the performance analysis on the medical image is the Choledoch dataset image [56]. Figure 10 shows the 2D and 3D histograms achieved for the medical hyperspectral image.

Table 7 Analysis of Cuprite hyperspectral image with the existing state-of-the-art methods

Cryptographic Method	Correlation Coefficients of Encrypted		Key Space	Entropy of Encrypted Image
	Vertical	Horizontal		
Logistic Map [15]	0.025141	0.017214	–	7.9924
Chebyshev Map [15]	0.032118	0.022741	–	7.9916
AES-256 [15]	0.010221	0.013217	1.2×10^{77}	7.9991
Hybrid Domain Encryption [15]	0.010012	0.012014	10^{28}	7.9991
Proposed Model	0.001732	0.001424	10^{96}	7.9995



(a) 2D Histogram of single spectra (b) 3D histogram of Original image (c) 3D histogram of Encrypted image

Fig. 10 2D and 3D Histograms of Medical hyperspectral images

Figure 10 depicts that the intensity values are more evenly scattered in the encrypted image, which depicts that there is a significant difference from the original image and the encrypted image is more secure to the statistical attacks.

Table 8 shows the different metric values obtained for the medical hyperspectral image using the proposed model.

From the table, it has been observed that the entropy value is more close to 8 for the encrypted image indicates that the pixel values of the encrypted image are random. Thus, the encrypted image does not leak any information regarding the original image. The attained NPCR and UACI values of the encrypted image show that the image is secure from the differential attacks. Also, the proposed model ensures that there is no data loss in the decrypted image using SSIM value. Thus, the proposed model achieves significant results in medical hyperspectral image.

6 Conclusion

This work presents a hybrid encryption model for hyperspectral image encryption and decryption for secure internet transmission. The SABES optimization technique is employed in the proposed hybrid model to optimize the chaotic maps' initial conditions and parameters. In the confusion process, random permutation is utilized to improve the randomness of image pixel values. The encryption module's diffusion phase employs multi-level chaotic maps and circular shift. For the satellite hyperspectral images, the proposed hybrid model has been applied. The proposed hybrid encryption model is validated using thirty-two satellite hyperspectral images with varying spatial and spectral values. For validation, the proposed model is evaluated against several types of tests (entropy, NPCR, and UACI). The proposed model achieves entropy values ranging from 7.9971 to 7.9995, which are extremely near to the necessary value of 8. Furthermore, the NPCR and UACI values show that the proposed model produces distinct encrypted images with one pixel modification. The pixel similarity and dissimilarity analysis shows that the encrypted images are quite different from the original images which ensures the information security

Table 8 Evaluation metrics for a comparison between original, encrypted, and decrypted medical images

Image Name	Entropy		NPCR	UACI	SSIM	
	Encrypted	Original			Original vs. Encrypted	Original vs. Decrypted
Choledoch Image	4.9136	7.9985	1	0.3334	0.0031	1.0000

from the intruders. Also, the proposed model has been shown to have no data loss after obtaining a decrypted image using the SSIM value. The randomness of the chaotic maps used in the propounded model has been proved using NIST test. The propounded model sufficiently tackles the problem of noise and data loss during the transmission. The key space analysis of the proposed model validates that the model has sufficient key space, which is higher than the 2^{128}. Also, the proposed model is very sensitive to the minor change (10^{-16}) in the key-values. The noise attacks and cropping attacks analysis prove that the proposed hybrid model has resistance against data loss during transmission. The comparison with the existing encryption methods proves that the proposed hybrid encryption model attains better results and needs less computation time for the encryption and decryption of the hyperspectral images. The findings show that the proposed hybrid model is stable, efficient, and safe against entropy, differential and brute force threats. The proposed model is also tested on medical hyperspectral images to demonstrate its utility in medical field data security.

Appendix A: Entropy, NPCR, UACI, and computational time result analysis

The proposed algorithm has been analyzed for the thirty-two hyperspectral images and the results of the performance metrics entropy, NPCR, UACI, Encryption and decryption time are enlisted in the Table 9.

Table 9 Result analysis of the proposed hybrid encryption model using entropy, NPCR, UACI, and execution time

Image Name	Dimension	Entropy		NPCR	UACI	Computational Time	
		Original	Encrypted			Encryption Time(s)	Decryption Time(s)
Indian Pines	145 × 145 × 220	3.8198	7.999	1	0.3334	1.8073	1.5557
Pavia University	610 × 340 × 103	3.6689	7.9987	1	0.3334	9.9777	6.9838
KSC	512 × 614 × 176	0.9017	7.9987	1	0.3334	24.7244	19.3515
Botswana	1476 × 256 × 145	4.0704	7.9986	1	0.3339	25.2113	18.3215
Salinas	512 × 217 × 224	3.6402	7.9988	1	0.3339	11.9996	8.64
China Change Dataset (T1)	420 × 140 × 154	3.5573	7.9971	1	0.4762	3.4446	2.7352
China Change Dataset (T2)	420 × 140 × 154	3.5573	7.9971	1	0.4763	3.4335	2.7145
USA Change Dataset (T1)	307 × 241 × 154	3.2549	7.9971	1	0.4832	6.2319	3.6783
USA Change Dataset (T2)	307 × 241 × 154	3.2549	7.9981	1	0.4832	6.2614	3.7985
Cuprite	512 × 614 × 224	3.4834	7.9995	1	0.4636	32.4078	22.8544
Washington DC	1280 × 307 × 191	5.2968	7.9982	1	0.4699	33.8633	24.952
Samson	95 × 95 × 156	3.382	7.9979	1	0.4914	0.8046	0.6186
Selftestrad	280 × 800 × 126	4.8118	7.9977	1	0.4819	12.9053	9.1638
Selftestrfl	280 × 800 × 126	3.8439	7.9969	1	0.4865	12.7957	9.3236
Urban 1	307 × 307 × 162	1.2511	7.9969	1	0.4975	7.8028	5.333
Urban 2	307 × 307 × 210	1.7464	7.9967	1	0.4974	9.2708	6.943
Jasper Ridge	100 × 100 × 198	3.6381	7.9984	1	0.4833	1.1368	0.6634
IEEE Grss CASI	349 × 1905 × 149	6.0019	7.998	1	0.4475	42.6387	30.7926

Table 9 (continued)

Image Name	Dimension	Entropy		NPCR	UACI	Computational Time	
		Original	Encrypted			Encryption Time(s)	Decryption Time(s)
Sandiego	256 × 256 × 189	4.2614	7.998	1	0.4736	7.4347	4.0221
Abu-Airport-1	100 × 100 × 205	3.9453	7.998	1	0.4907	1.0706	0.7159
Abu-Airport-2	100 × 100 × 205	3.9453	7.998	1	0.4906	0.9918	0.7337
Abu-Airport-3	100 × 100 × 205	3.9453	7.998	1	0.4921	1.0268	0.7653
Abu-Airport-4	100 × 100 × 291	3.9453	7.998	1	0.4883	1.3477	0.649
Abu-Beach-1	150 × 150 × 188	3.9453	7.998	1	0.4956	1.7498	1.5469
Abu-Beach-2	100 × 100 × 193	3.9453	7.998	1	0.4958	0.9804	0.7725
Abu-Beach-3	100 × 100 × 198	3.9453	7.998	1	0.491	1.034	0.7303
Abu-Beach-4	150 × 150 × 162	3.9453	7.998	1	0.4966	1.118	0.709
Abu-Urban-1	100 × 100 × 204	3.9453	7.998	1	0.4894	1.0932	0.7497
Abu-Urban-2	100 × 100 × 207	3.9453	7.998	1	0.4901	1.1	0.7465
Abu-Urban-3	100 × 100 × 191	3.9453	7.998	1	0.4904	1.0563	0.7145
Abu-Urban-4	100 × 100 × 205	3.9453	7.998	1	0.4916	0.9673	0.7032
Abu-Urban-5	100 × 100 × 205	3.9453	7.998	1	0.4912	1.0653	0.76

Appendix B: Result analysis of pixel similarity and dissimilarity metrics

The result analysis of the pixel similarity and dissimilarity metrics are enlisted in the Table 10.

Table 10 Result analysis of the proposed hybrid encryption model using pixel similarity and dissimilarity metrics

Image Name	Pixel Similarity Metrics					Pixel Dissimilarity Metrics		
	SSIM		SC	NCC	IF	PSNR	MSE	MAE
	Original vs. Encrypted	Original vs. Decrypted						
China Change Dataset (T1)	0.0016	1	0.0050	0.8320	-15.1273	5.2573	1.28E+09	55.0748
China Change Dataset (T2)	0.0015	1	0.0044	0.8173	-16.4295	5.2254	1.29E+09	47.6801
USA Change Dataset (T1)	0.0011	1	0.0025	0.7420	-25.1701	5.1033	1.33E+09	30.0554
USA Change Dataset (T2)	0.0010	1	0.0028	0.7526	-23.2919	5.0807	1.33E+09	26.7003
Cuprite	0.0017	1	0.0055	0.8503	-14.5231	5.2737	1.28E+09	59.9213
Washington DC	0.0013	1	0.0064	0.7404	-18.8051	5.1597	1.31E+09	70.133
Samson	0.0002	1	0.0001	0.7215	-183.9930	4.8179	1.42E+09	0.8818
Selftestrad	0.0008	1	0.0023	0.7010	-32.3726	5.0142	1.35E+09	253051
Selftestrfl	0.0006	1	0.0011	0.6893	-45.2044	4.9512	1.37E+09	11.8108

Table 10 (continued)

Image Name	Pixel Similarity Metrics					Pixel Dissimilarity Metrics		
	SSIM		SC	NCC	IF	PSNR	MSE	MAE
	Original vs. Encrypted	Original vs. Decrypted						
Urban 1	0.0001	1	0.0000	0.7368	-258.4358	4.8042	1.42E+09	0.3103
Urban 2	0.0001	1	0.0000	0.7308	-251.1369	4.8051	1.42E+09	0.3422
Jasper Ridge	0.0008	1	0.0017	0.7025	-33.6868	5.0051	1.36E+09	18.6386
IEEE Grss CASI	0.0033	1	0.0335	0.7246	-5.3072	5.8244	1.12E+09	365.46
Sandiego	0.0004	1	0.0050	0.7599	-16.8029	5.2132	1.29E+09	53.9365
Abu-Airport-1	0.0003	1	0.0006	0.8196	-61.9379	4.8994	1.39E+09	6.7252
Abu-Airport-2	0.0006	1	0.0006	0.8188	-62.1843	4.9008	1.39E+09	6.7666
Abu-Airport-3	0.0001	1	0.0004	0.8063	-82.1565	4.8740	1.38E+09	4.1557
Abu-Airport-4	0.0001	1	0.0010	0.8164	-44.5474	4.9544	1.37E+09	10.5948
Abu-Beach-1	0.0005	1	0.0002	0.3374	-196.6053	4.8159	1.42E+09	2.0876
Abu-Beach-2	0.0005	1	0.0001	0.4329	-247.0518	4.8083	1.42E+09	1.0551
Abu-Beach-3	0.0005	1	0.0007	0.7582	-60.8385	4.9052	1.39E+09	7.6471
Abu-Beach-4	0.0005	1	0.0000	0.5551	-44.4880	4.7926	1.42E+09	7.3437
Abu-Urban-1	0.0005	1	0.0008	0.7330	-55.0231	4.9215	1.38E+09	8.8691
Abu-Urban-2	0.0003	1	0.0008	0.6457	-57.7349	4.9124	1.39E+09	9.1232
Abu-Urban-3	0.0004	1	0.0008	0.7440	-56.4227	4.9180	1.38E+09	8.3969
Abu-Urban-4	0.0004	1	0.0005	0.7019	-71.9085	4.8871	1.39E+09	5.5975
Abu-Urban-5	0.0005	1	0.0006	0.7945	-67.9417	4.8947	1.39E+09	6.333

Data Availability The data that support the findings of this study are available on GIC Website [http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes], MicroImages Website [https://www.microimages.com/downloads/hyperspectral.html], Le Sun Webpage [http://lesun.weebly.com/hyperspectral-data-set.html], Remote Sensing Laboratory Website [https://rslab.ut.ac.ir/papers], Xudong Kang Webpage [http://xudongkang.weebly.com/data-sets.html], IEEE GRSS Data Website [https://hyperspectral.ee.uh.edu/?page_id=459#download], GitHub Website [https://github.com/huyanning/Hyperspectral-Anomaly-Detection], and MEDHSI Database Website [http://bio-hsi.ecnu.edu.cn/]

Declarations

Conflict of Interests The authors declare that they have no conflict of interest.

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